

Autocorrelated rate priors on phylogenies: a Thorne–Kishino–Painter relaxed clock for the gain–loss–duplication birth–death model

recount project
notation following CM09 SI and Csűrös 2021 arXiv

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Provenance and disclaimer. This document was drafted by Claude (Anthropic’s LLM, model `claude-opus-4-7[1m]`) in collaboration with the human author, as part of an interactive Claude Code session (<https://claude.com/claude-code>). The Thorne–Kishino–Painter autocorrelated-rate prior is standard (Thorne, Kishino & Painter, *MBE* 15:1647–1657, 1998); the per-edge gradient was finite-difference validated to relative error $\leq 10^{-6}$ against the analytical implementation in `native/src/recount.brownian.prior.c` and the Python reference in `validation/_shared.py: brownian_prior_and_grad_python`. Treat as “LLM-drafted, human-checked, FD-validated” engineering documentation rather than peer-reviewed manuscript content.

Abstract

We equip the `recount` maximum-a-posteriori fitter with a tree-structured prior that couples the rates of adjacent branches. The construction adapts the relaxed-clock prior of Thorne, Kishino & Painter (*MBE* 1998), placing a Brownian motion on the log-rate process as it walks down the phylogeny. We give the explicit log-density (Eq. (7)) and gradient (Eq. (8)), prove the prior is proper (Theorem 3) and that the resulting MAP objective is coercive (Theorem 5). The implementation is $\mathcal{O}(N)$ per evaluation and matches the analytical native C kernel to FD relative error $\leq 10^{-6}$.

1 Biological motivation

A gain–loss–duplication (GLD) phylogenetic fit estimates four quantities per branch of the species tree: a gain shape κ_v controlling *de novo* acquisition of new families, a duplication rate λ_v , a loss rate μ_v (conventionally fixed at one to set the timescale), and a branch length t_v . On a tree with N nodes that gives $\sim 3N$ free parameters. For the archaeal arc269 dataset ($N = 537$) this is $\sim 1,600$ rate parameters.

In every published reconstruction we are aware of these rates are treated as completely independent across branches. From an evolutionary-biology standpoint that is a strong, and usually unwarranted, assumption: the descendants of a recent common ancestor share an ecological niche, a mutational milieu, an effective population size, and (often) a metabolic strategy — all of which influence the rate of gene gain, loss, and duplication. A sudden, large jump in inferred gain rate from one branch to its child is biologically implausible most of the time, but the independent prior treats it as just as likely as no jump.

What is required is a prior that says “*adjacent branches should have similar rates, with gradually growing divergence over evolutionary time.*” This is exactly the structure introduced by Thorne, Kishino & Painter (1998, henceforth **TKP**) for molecular substitution rates: the log-rate is allowed to wander Brownian-style as one walks from the root down the tree, with variance growing with branch length. TKP’s prior was an early example of what is now called

a *relaxed clock*: it relaxes the strict-clock assumption (all branches share one rate) without collapsing to the unstructured independent-rate limit.

What this document does. We adapt TKP to GLD rates. Each axis (gain, duplication, length) gets its own independent Brownian process on the log-rate as it walks from the root through the tree. We work in the coordinates used by the optimiser (§2), give the explicit log-prior density (§4) and gradient (§5), prove that the resulting joint prior on rates is a full-rank Gaussian Markov random field, hence proper (§6.1), and prove that when combined with the GLD likelihood the MAP objective is coercive (§6.2). This last point is the key practical payoff: it rules out the rate-boundary asymptote analysed in the companion note `boundary_identifiability.tex`.

2 Setup and notation

We follow the notation of Csűrös 2021 (arXiv:2107.11440), with the survival-parameter convention of the CM09 supplement.

Tree. Let $\mathcal{T} = (V, E, R)$ be a rooted tree with node set V of cardinality N , root $R \in V$, leaf set $\mathcal{L} \subset V$, and edges directed from R outward. For non-root v we write $\text{pa}(v) \in V$ for its parent. Each edge $\text{pa}(v) \rightarrow v$ is identified with its child endpoint v .

Per-edge rates. On the incoming edge of each non-root node v we attach four parameters:

$$\kappa_v \geq 0, \quad \lambda_v \geq 0, \quad \mu_v > 0, \quad t_v \in (0, \infty). \quad (1)$$

κ_v is the immigration (Pólya shape) parameter for new gene families landing on the edge, λ_v the per-lineage duplication rate, μ_v the per-lineage loss rate, and t_v the time elapsed on the edge. Following CM09 we set $\mu_v \equiv 1$ (it absorbs the timescale), so the free per-edge parameters are $(\kappa_v, \lambda_v, t_v)$. The root R carries a gain parameter κ_R but no edge length.

The optimiser’s coordinates. Because rates are positive, the optimiser works on log-coordinates; for duplication we additionally apply the sub-critical Yule logit when activated. With N nodes (including root) define the packed vector

$$x = \left(\underbrace{\log \kappa_v}_{v \in V}, \underbrace{\theta_{\lambda,v}}_{v \neq R}, \underbrace{\log t_v}_{v \neq R} \right) \in \mathbb{R}^{3N-2}, \quad (2)$$

where

$$\theta_{\lambda,v} = \begin{cases} \text{logit}(\lambda_v / Q) & \text{(sub-critical Yule, } Q = 1 - 2^{-30}) \\ \log \lambda_v & \text{(unconstrained mode)} \end{cases} \quad (3)$$

matches the Csűrös Java convention (`is_duprate_bounded=true`). The gain axis includes the root; the duplication and length axes have $N - 1$ entries each (no root). To unify the three axes for the prior, write $x_v^{(a)}$ for the entry of x at node v on axis $a \in \{\kappa, \lambda, t\}$, with the convention that $x_R^{(\lambda)}$ and $x_R^{(t)}$ are identified with fixed reference centres (see §4 below).

Data. A gene-family profile is a vector $\boldsymbol{\xi} \in \mathbb{Z}_{\geq 0}^{\mathcal{L}}$ of integer copy counts at each leaf. The data is $\{\boldsymbol{\xi}_f\}_{f=1}^F$. The log-likelihood $\mathcal{L}(x) := \sum_f \log P(\boldsymbol{\xi}_f \mid x)$ is computed by the inside–outside recursion of CM09; this document does not modify it.

3 The prior: classical TKP and our v1 simplification

3.1 Classical TKP on log-rates

In its original form, TKP places a Gaussian on the log-rate *increment* between parent and child, with variance proportional to elapsed evolutionary time:

$$\log r_v - \log r_{\text{pa}(v)} \sim \mathcal{N}(0, \sigma_{\text{TKP}}^2 \cdot t_v). \quad (4)$$

This says: the log-rate process is a Brownian motion run along the tree, with diffusion coefficient σ_{TKP}^2 . Two sister branches with short edges between them and a recent common ancestor will have similar rates; two distant branches separated by long edges will be free to diverge.

3.2 Our v1: time-uniform Brownian

We adopt the structure of (4) with one simplification: drop the per-edge t_v weighting, so that every edge incurs the same variance σ_a^2 on each rate axis a . Formally,

$$\Delta_a(v) := x_v^{(a)} - x_{\text{pa}(v)}^{(a)} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_a^2) \quad \text{for every } v \neq R, \quad a \in \{\kappa, \lambda, t\}. \quad (5)$$

This is mathematically a Gaussian process on the tree with covariance function $\text{Cov}(x_u, x_v) = \sigma_a^2 \cdot d(u, v)$ where $d(u, v)$ is the *topological* (edge-count) distance to the most recent common ancestor, rather than the metric distance. Biologically it treats every branch as the same “unit of evolutionary opportunity” to revise the rate, irrespective of its t_v .

Remark 1 (Why drop the t_v weight?). The classical TKP weight t_v couples the prior to the very quantity (t_v itself) that the optimiser is solving for, introducing a non-trivial extra chain-rule term whenever t_v moves. The time-uniform variant produces an $\mathcal{O}(N)$ kernel with no t -coupling, and empirically captures the same first-order effect (shrinkage between adjacent branches). The classical form is left for a v2 implementation; see §9.

3.3 Anchoring the prior at the root

A Brownian motion has no global scale: shifting every $x_v^{(a)}$ by a common constant leaves all the increments unchanged. To pin the absolute scale we add a wide independent Gaussian *root anchor*

$$\log \kappa_R \sim \mathcal{N}(\mu_R^{(\kappa)}, \sigma_R^2), \quad (6)$$

where by convention we use $\mu_R^{(a)}$ for the prior centre on axis a . On the duplication and length axes the root does not have an optimisable rate (the root has no incoming edge), so the root values are fixed at the prior centres $\mu_R^{(\lambda)}, \mu_R^{(t)}$ and “ $\Delta_a(v)$ for $\text{pa}(v) = R$ ” is interpreted with these fixed values plugged in.

4 Log-density

Combining (5) and (6), the joint log-prior is

$$\log p_{\text{B}}(x) = -\frac{1}{2} \sum_{a \in \{\kappa, \lambda, t\}} \sum_{v \neq R} \frac{\Delta_a(v)^2}{\sigma_a^2} - \frac{1}{2} \frac{(\log \kappa_R - \mu_R^{(\kappa)})^2}{\sigma_R^2} + \text{const.} \quad (7)$$

The constant absorbs the normalising factors of the Gaussians; under the v1 time-uniform variance they do not depend on x .

Worth pausing on. Equation (7) is just a sum of squared log-rate increments along the tree — one per edge, per axis, plus one root-anchor term on the gain axis. There are no off-tree interactions, no quadratic couplings between non-adjacent branches. This is what makes the gradient and the inference $\mathcal{O}(N)$.

5 Gradient

The gradient is obtained by direct differentiation of (7). Each edge contribution $-\Delta_a(v)^2/(2\sigma_a^2)$ depends on the child entry $x_v^{(a)}$ and on the parent entry $x_{\text{pa}(v)}^{(a)}$ (the latter only when $\text{pa}(v)$ is itself not the root, or on the gain axis where the root is in x). The closed form is

$$\frac{\partial \log p_B}{\partial x_v^{(a)}} = -\frac{\Delta_a(v)}{\sigma_a^2} \quad (\text{contribution from } v \text{ as a child}), \quad (8)$$

$$\frac{\partial \log p_B}{\partial x_{\text{pa}(v)}^{(a)}} += +\frac{\Delta_a(v)}{\sigma_a^2} \quad (\text{contribution from } \text{pa}(v) \text{ as a parent of } v), \quad (9)$$

$$\frac{\partial \log p_B}{\partial \log \kappa_R} += -\frac{\log \kappa_R - \mu_R^{(\kappa)}}{\sigma_R^2} \quad (\text{root anchor}). \quad (10)$$

The “+=” notation captures the scatter-add semantics: the parent of any internal node u accumulates contributions from all of u ’s children $v \in \text{ch}(u)$. The implementation is a single pass over the $N - 1$ tree edges with $\mathcal{O}(1)$ work per edge per axis.

Proposition 2 (Computational complexity). *The log-prior (7) and its full gradient (8)–(10) cost $\mathcal{O}(N)$ floating-point operations.*

Proof. One pass over the $N - 1$ tree edges with three scalar additions per edge per axis, plus a single root-anchor term. $\square$

6 Theoretical properties

6.1 Propriety: the prior is a full-rank GMRF

We show that the joint prior on the packed vector x defined by (7) is a non-degenerate (full-rank) Gaussian distribution. Equivalently, it integrates to a finite constant, so it is a *proper* prior in the Bayesian sense.

Theorem 3 (Propriety). *Fix $\sigma_a > 0$ for $a \in \{\kappa, \lambda, t\}$ and $\sigma_R > 0$. Under the conventions of §2–§4, the prior density (7) arises from a Gaussian distribution on $x \in \mathbb{R}^{3N-2}$ with positive-definite precision matrix*

$$\Sigma^{-1} = \text{blkdiag}(\Lambda_\kappa, \Lambda_\lambda, \Lambda_t), \quad (11)$$

where on the gain axis $\Lambda_\kappa = (1/\sigma_\kappa^2) L_{\mathcal{T}} + (1/\sigma_R^2) e_R e_R^\top$ with $L_{\mathcal{T}}$ the unsigned graph Laplacian of $\mathcal{T}$, and on the duplication and length axes $\Lambda_a = (1/\sigma_a^2) L_{\mathcal{T}}^\partial$ is the Dirichlet (root-clamped) tree Laplacian. In particular each block is positive-definite and the joint prior integrates to a finite constant.

Proof. We argue axis by axis. Fix an axis a . The terms in (7) that depend on the a -axis entries of x form a quadratic form $Q_a(x) = (1/2\sigma_a^2) \sum_{v \neq R} \Delta_a(v)^2 + (1/2\sigma_R^2)(x_R^{(a)} - \mu_R^{(a)})^2$ on the gain axis). We must show Q_a is positive-definite.

Gain axis ($a = \kappa$). For every edge $\text{pa}(v) \rightarrow v$, the term $\Delta_\kappa(v)^2 = (x_v^{(\kappa)} - x_{\text{pa}(v)}^{(\kappa)})^2$ contributes $+1/\sigma_\kappa^2$ to the (v, v) and $(\text{pa}(v), \text{pa}(v))$ diagonals and $-1/\sigma_\kappa^2$ to the off-diagonals $(v, \text{pa}(v))$ and $(\text{pa}(v), v)$. Summing over edges gives exactly the unsigned graph Laplacian $L_{\mathcal{T}} = D - A$ scaled

by $1/\sigma_\kappa^2$, where D is the degree matrix and A the adjacency matrix. This $L_{\mathcal{T}}$ has nullspace spanned by the all-ones vector $\mathbf{1}$ (Brownian motion has a free additive constant). The root anchor $(1/\sigma_R^2)(x_R^{(\kappa)} - \mu_R^{(\kappa)})^2$ contributes $+1/\sigma_R^2$ to the (R, R) diagonal, which removes that nullspace direction: any non-zero $y \in \mathbb{R}^N$ either has $y \not\propto \mathbf{1}$ (in which case $L_{\mathcal{T}}$ already gives strictly positive contribution) or $y = c\mathbf{1}$ with $c \neq 0$ (in which case the root-anchor term gives $c^2/\sigma_R^2 > 0$). Hence $\Lambda_\kappa \succ 0$.

Duplication and length axes ($a \in \{\lambda, t\}$). On these axes the root entry is not in x but is held fixed at the prior centre $\mu_R^{(a)}$. The edge-increment terms involving the root, $(x_v^{(a)} - \mu_R^{(a)})^2$ for $v \in \text{ch}(R)$, then contribute $+1/\sigma_a^2$ to the (v, v) diagonal but no off-diagonal. This is the standard Dirichlet condition: the “root-clamped” tree Laplacian $L_{\mathcal{T}}^\partial$ obtained from $L_{\mathcal{T}}$ by deleting the root row/column has full rank (a connected graph minus one boundary node has full-rank Laplacian). Hence $\Lambda_a \succ 0$.

The three axes do not interact in (7), so the full precision is block-diagonal and positive-definite. The joint density is a non-degenerate Gaussian and integrates to $(2\pi)^{(3N-2)/2} \det(\Sigma)^{1/2}$, finite. $\square$

Remark 4 (Why the root anchor on κ is necessary). Without the root anchor, Λ_κ is the bare tree Laplacian $L_{\mathcal{T}}$, which is rank-deficient: it admits a one-dimensional nullspace corresponding to a common additive shift of all $\log \kappa_v$. Biologically that is the “slide the entire gain rate up and down” direction — the data *can* pin it (a high gain rate predicts many copies; a low gain rate predicts few), but the *prior* cannot, and an improper prior on a direction the data also struggles to constrain would propagate to a non-identifiable posterior. The wide anchor $\sigma_R = 5$ contributes very little except in the marginal axis the data cannot fix; it is the minimum surgery to make the prior proper.

6.2 Coercivity of the MAP objective

The point of using the prior at all is that the GLD likelihood $\mathcal{L}(x)$ is not bounded above on $\mathbb{R}^{3N-2}$: at certain boundary configurations the supremum is finite but unattained on $\mathbb{R}^{3N-2}$ (see the companion note `boundary_identifiability.tex` for the Pólya–Poisson asymptote analysis). The Brownian prior penalises those configurations, and combined with the likelihood gives a coercive objective.

Theorem 5 (Coercivity of the Brownian-MAP objective). *For any finite $\sigma_a, \sigma_R > 0$, the objective*

$$\mathcal{L}^{\text{MAP,B}}(x) := \mathcal{L}(x) - F \log(1 - L(0; x)) + \log p_B(x) \quad (12)$$

is coercive on $\mathbb{R}^{3N-2}$: $\mathcal{L}^{\text{MAP,B}}(x_n) \rightarrow -\infty$ whenever $\|x_n\| \rightarrow \infty$. Its supremum is therefore attained at an interior point.

Proof. The likelihood term $\mathcal{L}(x) = \sum_f \log L_f(x)$ is non-positive ($L_f \in (0, 1]$ for every family). The conditional-observability correction $-F \log(1 - L(0; x))$ is positive but bounded above by $F \log(\liminf_{x \in \mathbb{R}^{3N-2}} (1 - L(0; x)))^{-1}$, which is finite as long as the family-size process is non-degenerate (it is, on any tree). Hence the first two terms are bounded above.

The third term $\log p_B(x)$ is a quadratic form $-\frac{1}{2}(x - \mu)^\top \Sigma^{-1}(x - \mu) + \text{const}$, where Σ^{-1} is positive-definite by Theorem 3 and μ is the prior mean. Hence $\log p_B(x) \leq -\frac{1}{2} \lambda_{\min}(\Sigma^{-1}) \|x - \mu\|^2 + \text{const}$, where $\lambda_{\min}(\Sigma^{-1}) > 0$. As $\|x\| \rightarrow \infty$ this tends to $-\infty$ at quadratic rate, dominating any finite bounded likelihood term, so $\mathcal{L}^{\text{MAP,B}}(x) \rightarrow -\infty$.

Coercivity together with continuity of $\mathcal{L}^{\text{MAP,B}}$ (continuity of $\mathcal{L}$ is standard from the smoothness of the survival map of CM09; the prior is a smooth quadratic; $L(0; \cdot)$ is smooth by the Pn-tensor recursion of Csürös 2026 SI) implies that every super-level set $\{x : \mathcal{L}^{\text{MAP,B}}(x) \geq c\}$ is non-empty for some c and bounded by coercivity, hence compact, on which a continuous function attains its supremum. $\square$

7 Default hyperparameters and biological calibration

parameter	default	biological meaning
σ_κ	1.0	one e -fold per edge in gain shape
σ_λ	1.0	one logit-unit per edge in duplication
σ_t	1.0	one e -fold per edge in branch length
σ_R	5.0	deliberately wide; identifiability only
$\mu_R^{(\kappa)}$	$\log 0.1$	matches <code>--mu-gain</code>
$\mu_R^{(\lambda)}$	$\text{logit}(0.5) = 0$	matches <code>--mu-dup</code>
$\mu_R^{(t)}$	$\log 1 = 0$	matches <code>--mu-length</code>

Table 1: Default hyperparameters. “One e -fold per edge” means the 68 % Brownian-walk prior interval after one edge step on a given axis is roughly ± 1 on the log-scale, i.e. multiplicative factors in $[e^{-1}, e^1] \approx [0.37, 2.72]$. Across a 20-edge-deep tree, the cumulative prior credible interval on log-rate is then roughly $\pm\sqrt{20} \approx \pm 4.5$, i.e. $\sim 90\times$ either direction — wide enough not to override data, tight enough to forbid pathological boundary excursions of $\sim 10^{10}\times$.

What “ $\sigma_a = 1$ ” is asking biologically. With $\sigma_a = 1$, two sister branches off a recent common ancestor are expected to have log-rates within ± 2 ($\sim 95\%$ prior CI under a one-step Brownian) — i.e. their rates within a factor of ~ 7 . That matches the kind of within-clade rate heterogeneity reported in molecular-clock studies (where the relaxed-clock literature finds factors of 3–10 across closely related taxa). Tightening to $\sigma_a = 0.5$ enforces visibly more smoothing; loosening to $\sigma_a \gg 1$ recovers the independent-prior limit.

CLI coordinate convention. The CLI flag `--mu-dup` in `validation/map.py` is specified by the user in *log-rate units* (default $\log 0.5 \approx -0.693$, i.e. “prior centre at $\lambda = 0.5$ ”), not in the optimiser’s logit-coord units. When the sub-critical constraint is active the optimiser acts on $\theta_{\lambda,v} = \text{logit}(\lambda_v/Q)$, so the prior centre $\mu_R^{(\lambda)}$ appearing in (7) must also be in logit-coords. The translation happens once at the boundary of `fit_dataset` via `_mu_dup_for_optimizer`; each summary JSON records both the user-passed log-rate value and the translated logit-coord value (`mu_dup_in_optimizer_coord`).

8 Implementation

- **Native C:** `native/src/recount_brownian_prior.c`. Public symbol `recount_brownian_prior_and_grad` single $\mathcal{O}(N)$ loop over edges, no allocation, no threading (work is too small to benefit from `dispatch_apply`).
- **ctypes wrapper:** `recount.native_backend.brownian_prior_native`, accepting the parent array, root index, packed x , the four sigmas, and the three prior centres; returning $(\log p_B, \nabla \log p_B)$.
- **Python reference:** `validation._shared._brownian_prior_and_grad_python` uses `numpy.add.at` for the scatter-add and is kept for finite-difference cross-checking in `tests/test_brownian_prior.py`.
- **Integration with the existing MAP factory:** `make_objgrad_map_brownian` in `validation/_shared.py` subtracts $\partial \log p_B / \partial x$ from the negative-LL gradient produced by `_gradient_raw_native`; the LL chain rule is unchanged.

- **Validation:** analytical gradient matches central finite differences to relative error $\leq 10^{-6}$ on 30 random optimiser positions on `dpann80` ($N = 159$); the full-objective gradient through `make_objgrad_map_brownian` matches FD to the same tolerance. See `tests/test_brownian_prior` for the four-test suite.

9 Out of scope for v1, deferred to v2

1. **TKP-classic per-edge t_v weighting** ((4), the proper diffusion form). Requires extra chain-rule terms since t_v is itself in the optimiser. *Consequence of the v1 simplification:* very long branches are penalised as tightly as very short ones for a 1-unit log-rate change. On real phylogenies with heterogeneous branch lengths, the v1 prior systematically over-smooths short branches and under-smooths long branches relative to the proper TKP form.
2. **Empirical Bayes / REML estimation of the σ_a .** V1 fixes the three sigmas from the CLI; an outer EM or REML loop is a v2 enhancement.
3. **Composition with the K -category LogisticShift mixture** (`logistic_shift_gradient.tex`). The current mixture fit applies an independent Gaussian prior on the base rates; replacing it with the Brownian prior would compose family-level heterogeneity with branch-level smoothing.
4. **Per-branch discrete rate-class HMM.** A separate model class with its own forward-backward, beyond the scope of the Brownian-prior framework.

10 Related work

- **Thorne, Kishino & Painter (1998)**, *Estimating the rate of evolution of the rate of molecular evolution*, MBE 15:1647–1657. The original autocorrelated-rate prior; we adapt the structure unchanged.
- **Drummond, Ho, Phillips & Rambaut (2006)**, *Relaxed phylogenetics and dating with confidence*, PLoS Biology 4:e88. The uncorrelated-lognormal alternative ($\log r_v$ iid from a global distribution, ignoring tree structure). The recount independent prior is closer to this unstructured form; the Brownian prior described here is its TKP autocorrelated counterpart.
- **Csűrös & Miklós (2009)**, *Streamlining and large ancestral genomes in Archaea inferred with a phylogenetic birth-and-death model*, MBE 26:2087–2095. The GLD model whose per-node rates we are placing a prior on. Csűrös’ Java reference (`count.model.MLDistribution`) does plain maximum-likelihood optimisation of the rates: no Bayesian prior on the per-node $(\kappa_v, \lambda_v, t_v)$. The only “prior” in Csűrös’ code is the *root prior* on the copy count at the root, which is part of the model itself, not a regulariser on the rate parameters; and the “MAP” language in Csűrös 2026 PNAS refers to maximum-a-posteriori family-presence inference (a per-family binary classifier on the inferred copy counts at each internal node), not MAP rate estimation. The independent per-node log-Normal prior in `validation/map.py` as the `--prior independent` mode, and the Brownian prior described here as `--prior brownian`, are both additions by the present codebase.
- **Csűrös (2021)**, arXiv:2107.11440. Analytical gradient of the GLD likelihood via the conserved-copy decomposition. Provides the per-rate gradients used in our $\mathcal{L}(x)$ pipeline.